	Course Name	**Lighting Technology**	
	Student Name	**Mirjabbar Badalov**	
	Year	**2025-2026**	Study group
	Collaborator		Measurement Date 06/11/2025

Task name: Measurement of photometric quantities with a spectroradiometer

1 Measurement Task

This lab exercise focuses on measuring photometric properties with a spectroradiometer. Its purpose is to illustrate how to determine radiance, luminance, colour coordinates, and correlated color temperature using the instrument.

2 Task Analysis

Light is a type of electromagnetic radiation that the human eye can perceive. The visible spectrum extends from violet to red, and our eyes are most responsive to this range in daylight conditions. In 1923, scientists developed a standard sensitivity curve, known as V(λ), based on the behaviour of cone cells in the eye, and this standard was officially adopted in 1924.

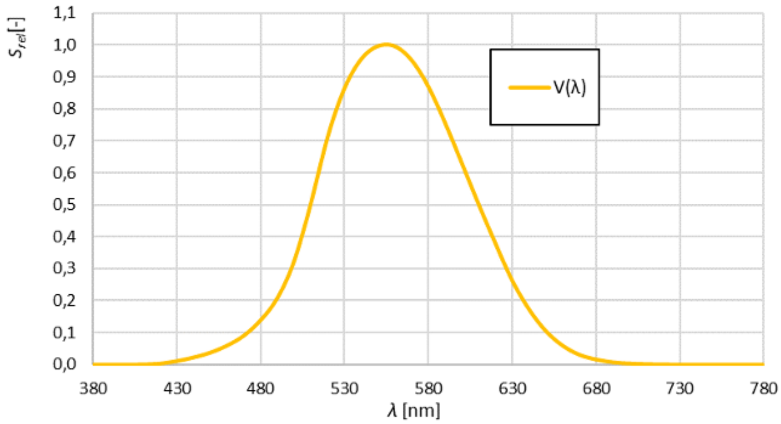


Figure 1The spectral sensitivity of human eye

When discussing light in everyday contexts, we use photometry—a field that focuses on how light impacts human vision. Rather than dealing with energy-based measures like radiant flux, photometry uses quantities that reflect how the eye responds to different wavelengths. These include luminous flux (the amount of visible light), luminous intensity (perceived brightness), and luminance (brightness across a surface), among other related terms. In short, lighting technology is concerned with how light is perceived, not just its physical energy.

The CS-1000 expresses the results of spectral measurements not in watts, but in units of $W \cdot sr^{-1} \cdot m^{-2} \cdot nm^{-1}$, which indicate spectral radiance density (Leλ). This corresponds to integral radiance (Le), measured in $W \cdot sr^{-1} \cdot m^{-2}$. Likewise, the photometric quantity known as luminance (L or Lv) can be calculated, and its standard unit is candela per square meter (cd/m^2).

By using the spectral radiance density (Leλ), color coordinates can be determined. Trichromatic color systems—based on the concept of additive color mixing—show that any color can be produced by combining three independent primary colors. In the CIE 1931 color model, real colors are represented through a mixture of three imaginary primary lights: X, Y, and Z. Changing the amounts of X and Z alters the hue without affecting the total brightness of the stimulus. These three components can then be converted into the normalized trichromatic coordinates x, y, and z, using the CIE 1931 color-matching functions $\overline{x}(\lambda)$, $\overline{y}(\lambda)$, and $\overline{z}(\lambda)$, which are provided in standardized reference tables. Plotting the x and y coordinates in the CIE 1931 chromaticity diagram makes it possible to visualize the perceived color corresponding to the measured spectral radiance.

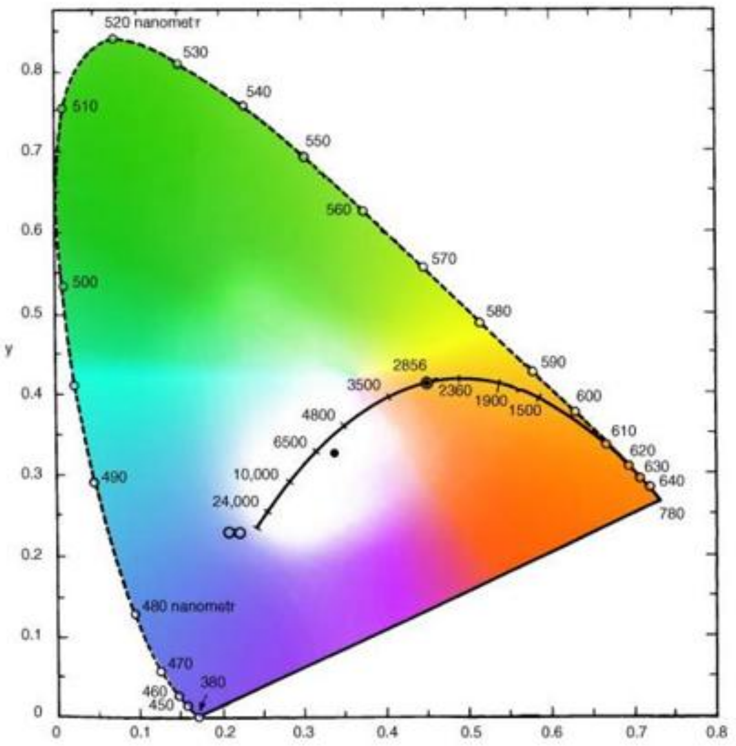


Figure 2 Trichromatic colour space CIE 1931 (XYZ)

Correlated color temperature (Tcp) describes the color appearance of white light and is expressed in kelvin (K). A higher temperature corresponds to a cooler, more bluish tone. For instance, incandescent lamps at about 2700 K give off a warm yellowish light, neutral white is roughly 4000 K, and cool daylight-like white is around 6500 K. Tcp can be determined using the Robertson method or by referring to a correlated color temperature chart. In Figure 3, the Planckian locus within the CIE 1931 color space illustrates how Tcp can be identified from the x and y color coordinates.

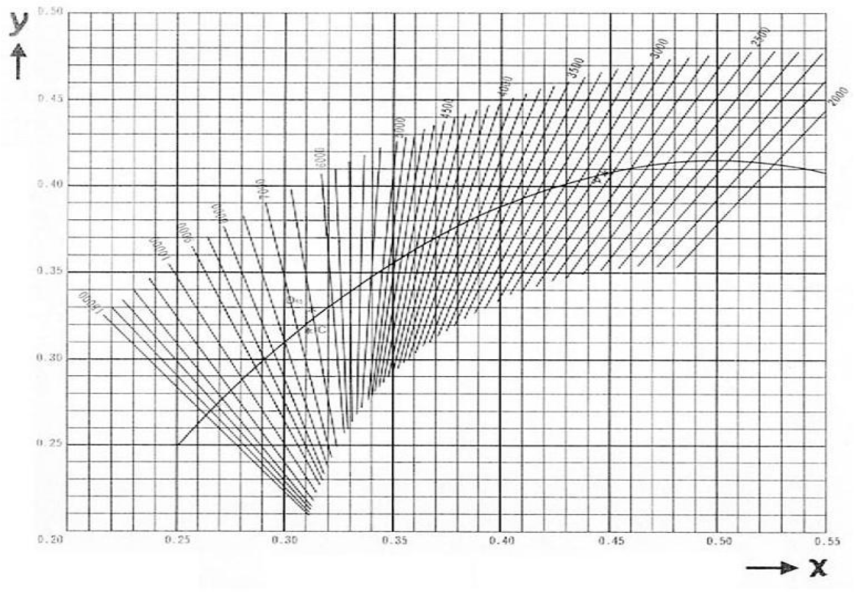


Figure 3Diagram of correlated colour temperature in CIE 1931 colour space

3 Apparatus and equipment used.

- Spectroradiometer KM-CS1000A.
- Computer.
- CS-S10w program

4 Measurement procedure

1. Ensure the spectroradiometer is properly connected to the computer, power on all required equipment, and remove the lens cap.
2. Open the CS-S10w software and create a new project (File → New).
3. Load the "MSVT-Mereni" template using File → Template → Load Template.
4. Initiate communication between the spectroradiometer and the PC by pressing F5.
5. Turn on the selected light source in the test box and allow it to stabilize.
6. Press F4, enter a description of the light source, and begin the measurement.
7. Switch off the current light source, point the spectroradiometer at the next one, and continue.
8. Repeat steps 5–7 for all prepared light sources.
9. Select the measured data by clicking the grey box in the top-left corner of the table, copy it, and paste it into an external file (e.g., Excel).
10. Power down all equipment and replace the lens cap.
11. Compute the necessary quantities, compare them with the measured values, assess any differences, and complete the report.

5 Tabular and graphical results

Table of measured and calculated values:

To calculate the Radiance, we used the equation given by:

$$L_{e\lambda} = \int_{380}^{780} L_{e\lambda}(\lambda) \cdot d\lambda$$

$$L_{e\lambda} = \sum_{i=380}^{780} L_{e\lambda}(\lambda_i).d\lambda$$

Since the step is 1 nm in the measured values, delta lambda in this calculation will be $d\lambda = 1$

$$L_v = 683 \int_{380}^{780} L_{e\lambda}(\lambda) \cdot V(\lambda) \cdot d\lambda$$

Where $V(\lambda)$ is the spectral sensitivity of the standard observer and can be found in standardized tables. The maximum luminous efficiency is defined for 555 nm monochromatic radiance with a constant 683 lm/W.

The following table represents the measured values compared to the calculated values:

Table 1Measured and calculated values of Le and Lv

Lamp Name	Le	Le_cal	Lv	Lv_cal
BLUE LIGHT BULB	19.20	19.16	369.62	368.82
YELLOW BULB	59.90	59.94	8513.65	8517,01
ORANGE BULB	36.60	36.55	4496.18	4494.14
RED LIGHT BULB	5.95	5.95	58.68	61.47
TRANSPERENT BULB	261.00	260.85	38931.03	38931
OSRAM 60WATT LIGHT BULB	343.00	343.03	52251.85	52242,67
PHILIPS 60 WATT LIGHT BULB	284.00	283.84	41625.08	41628,85
PHILIPS 45WATT LIGHT BULB	262.00	261.55	39847.37	39853,05
OSRAM 14WATT LIGHT BULB	287.00	286.73	44855.49	44845,78
LED TWIST FLAIR LIGHT BULB	109.00	109.32	35205.53	35194,99
LED TESLA 6 WATT LIGHT BULB	63.40	63.36	20376.96	20373,89
LED OSRAM 2 WATT LIGHT BULB	19.80	19.79	7138.81	7137,35
LED OSRAM 2.3 WATT LIGHT BULB	29.50	29.50	9163.37	9159,03
VITAE-Z 9 LIGHT BULB M1	1.60	1.60	531.98	532,74
VITAE-Z 9 LIGHT BULB M2	31.70	31.65	2940.98	2943,73
VITAE-Z9 LIGHT BULB M3	36.00	36.05	8351.65	8353,09
ATTRALUX LED LIGHT BULB 10 W	18.50	18.52	6005.08	6017,23
OSRAM FLOURESCENT LAMP 20 W	19.60	19.59	6951.56	7335,91
STICK OSRAM LAMP 20W	21.80	21.79	7760.01	8025,25
MEGAMAN FL 15 W	35.20	35.18	11635.64	11822,73
ENDURA PHILIPS LED 12W	182.00	181.72	54688.06	54694,64
PANASONIC LED 4.4 W	29.90	29.92	9339.76	9329,78
FLOURESCENT PHILIPS LUXLINE 18 W 83	25.10	25.11	8770.73	8715,08

From the table, it is evident that the calculated **Le** matches the measured value exactly. However, the **Lv** values differ significantly. Although changing the wavelength step size distorts the spectrum and affects the outcome, this alone does not fully account for such a large discrepancy.

It is also notable that the values of **V(λ)** are all less than 1. Consequently, when **Le(λ)** is multiplied by these values, the resulting product is naturally smaller than **Le(λ)**.

Correlated colour temperature from the diagram in Fig. 3:

Data Name	x	y	T
BLUE LIGHT BULB	0.2232	0.275	------
YELLOW BULB	0.5153	0.4459	2337
ORANGE BULB	0.5337	0.4191	------
RED LIGHT BULB	0.6998	0.2983	------
TRANSPERENT BULB	0.4581	0.4107	2731
OSRAM 60WATT LIGHT BULB	0.4554	0.4102	2764
PHILIPS 60 WATT LIGHT BULB	0.4642	0.4125	2664
PHILIPS 45WATT LIGHT BULB	0.4526	0.4084	2790
OSRAM 14WATT LIGHT BULB	0.4484	0.4086	2853
LED TWIST FLAIR LIGHT BULB	0.4788	0.4234	2559
LED TESLA 6 WATT LIGHT BULB	0.4496	0.3962	2732
LED OSRAM 2 WATT LIGHT BULB	0.4473	0.4135	2910
LED OSRAM 2.3 WATT LIGHT BULB	0.3215	0.3241	6050
VITAE-Z 9 LIGHT BULB M1	0.6319	0.3676	------
VITAE-Z 9 LIGHT BULB M2	0.5692	0.4104	------
VITAE-Z9 LIGHT BULB M3	0.5326	0.4208	------
ATTRALUX LED LIGHT BULB 10 W	0.4525	0.4066	2776
OSRAM FLOURESCENT LAMP 20 W	0.4471	0.4151	2927
STICK OSRAM LAMP 20W	0.4586	0.4158	2765
MEGAMAN FL 15 W	0.3842	0.3761	3892
ENDURA PHILIPS LED 12W	0.4526	0.406	2769
PANASONIC LED 4.4 W	0.4667	0.4164	2661
FLOURESCENT PHILIPS LUXLINE 18 W 83	0.4281	0.3983	3105

Calculation of the trichromatic components In the next table the trichromatic components X, Y calculated with the equation given by:

$$X = \int_{380}^{780} L_{e\lambda}(\lambda) \cdot \bar{x}(\lambda) \cdot d\lambda$$

$$Y = \int_{380}^{780} L_{e\lambda}(\lambda) \cdot \bar{y}(\lambda) \cdot d\lambda$$

$$\mathrm{X} = \sum_{\lambda=380}^{780} L_{e\lambda}(\lambda_i).\bar{x}(\lambda_i).d\lambda$$

$$\mathrm{Y} = \sum_{\lambda=380}^{780} L_{e\lambda}(\lambda_i).\bar{y}(\lambda_i).d\lambda$$

In this calculation we used the measured values given by a step of 1nm and the values of x, y which are given by a step of 5nm so we had to sort the measured data to match the rest of values in this case we take $d\lambda$=5 nm.

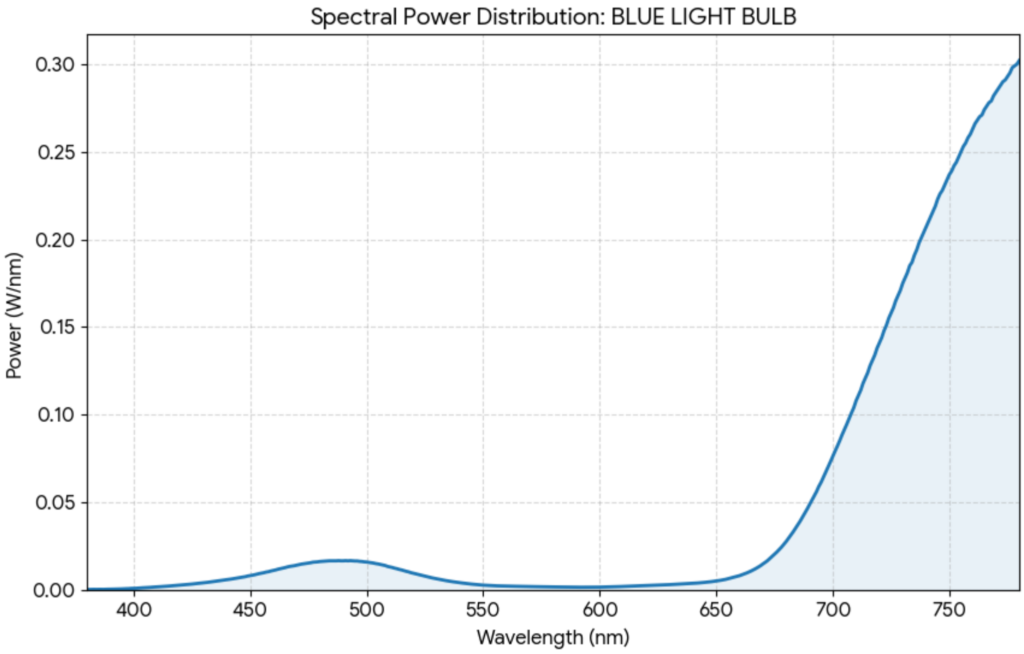
Then the trichromatic coordinates are calculated with the equation given by:

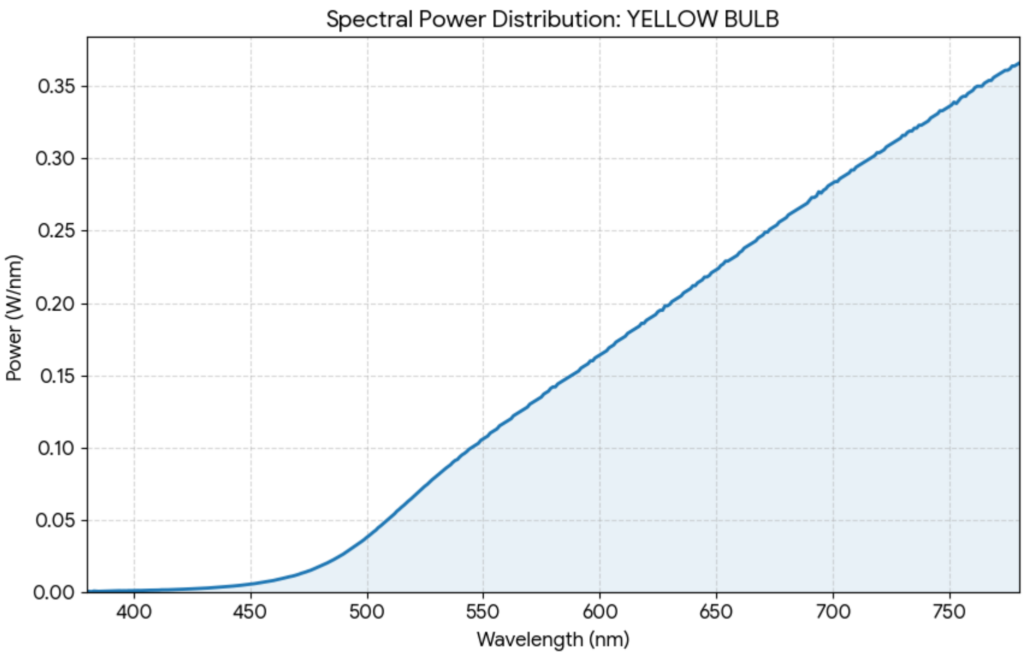
$$x = \frac{X}{X+Y}; \qquad y = \frac{Y}{X+Y}$$

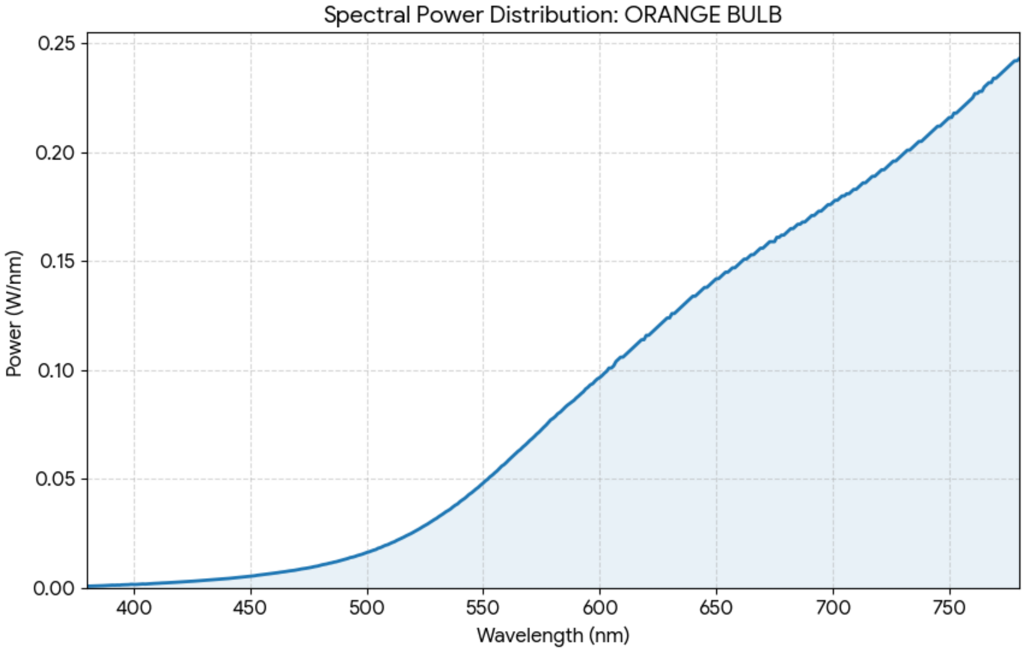
Lamp Name	Calculated X	Calculated Y	Calculated x	Calculated y	Measured x	Measured y
BLUE LIGHT BULB	0.6178	0.2867	0.3577	0.166	0.2232	0.275
YELLOW BULB	2.4221	9.0667	0.1858	0.6955	0.5153	0.4459
ORANGE BULB	1.1513	4.9218	0.1666	0.7124	0.5337	0.4191
RED LIGHT BULB	0.0017	0.0731	0.023	0.9685	0.6998	0.2983
TRANSPERENT BULB	16.8194	40.2672	0.2282	0.5464	0.4581	0.4107
OSRAM 60WATT LIGHT BULB	22.8947	53.9433	0.2299	0.5416	0.4554	0.4102
PHILIPS 60 WATT LIGHT BULB	17.4297	43.2205	0.2249	0.5576	0.4642	0.4125
PHILIPS 45WATT LIGHT BULB	17.6534	41.1124	0.2306	0.537	0.4526	0.4084
OSRAM 14WATT LIGHT BULB	20.3053	46.0842	0.2335	0.53	0.4484	0.4086
LED TWIST FLAIR LIGHT BULB	12.5485	37.3607	0.2048	0.6098	0.4788	0.4234
LED TESLA 6 WATT LIGHT BULB	9.2019	21.4385	0.2254	0.5252	0.4496	0.3962
LED OSRAM 2 WATT LIGHT BULB	2.7893	7.5749	0.2088	0.567	0.4473	0.4135
LED OSRAM 2.3 WATT LIGHT BULB	6.4435	8.8978	0.2562	0.3537	0.3215	0.3241
VITAE-Z 9 LIGHT BULB M1	0.0276	0.659	0.0399	0.9537	0.6319	0.3676
VITAE-Z 9 LIGHT BULB M2	0.6549	3.2499	0.1531	0.7597	0.5692	0.4104
VITAE-Z9 LIGHT BULB M3	2.3457	9.0484	0.1794	0.692	0.5326	0.4208
ATTRALUX LED LIGHT BULB 10 W	2.691	6.1978	0.2315	0.5332	0.4525	0.4066
OSRAM FLOURESCENT LAMP 20 W	2.5938	7.4835	0.2047	0.5907	0.4471	0.4151
STICK OSRAM LAMP 20W	2.7821	8.2502	0.2026	0.6007	0.4586	0.4158
MEGAMAN FL 15 W	6.343	11.6257	0.2456	0.4501	0.3842	0.3761
ENDURA PHILIPS LED 12W	22.0872	57.4812	0.2142	0.5576	0.4526	0.406
PANASONIC LED 4.4 W	3.4494	9.8488	0.207	0.5909	0.4667	0.4164
FLOURESCENT PHILIPS LUXLINE 18 W 830	3.8147	8.8674	0.2264	0.5263	0.4281	0.3983

We can see that the values are slightly different due to the distortion of the spectre of light after sorting the data.

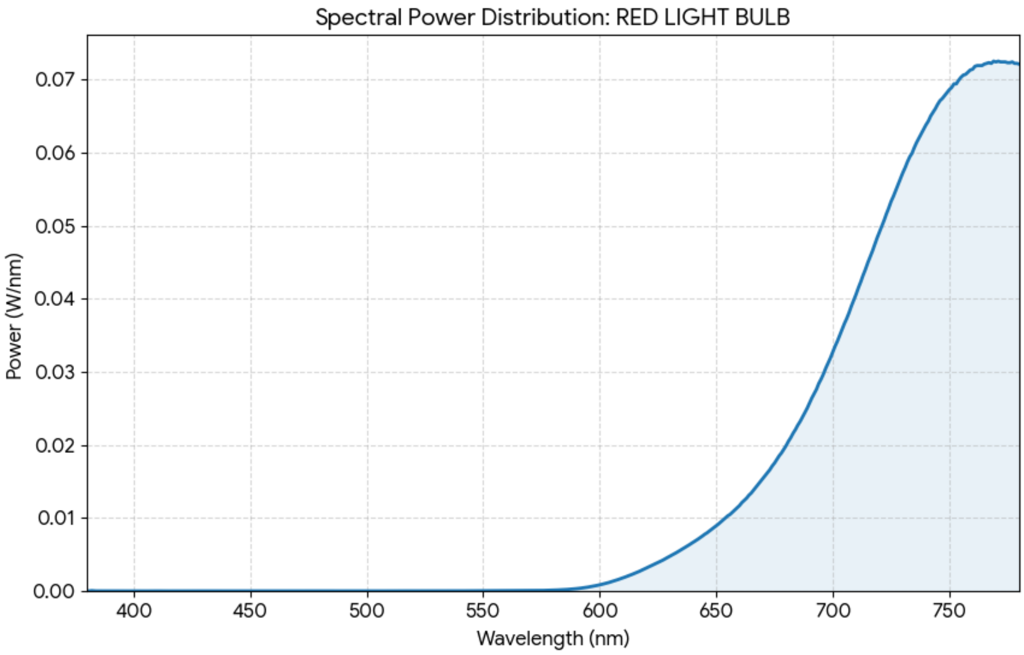
The measured spectra of the light sources



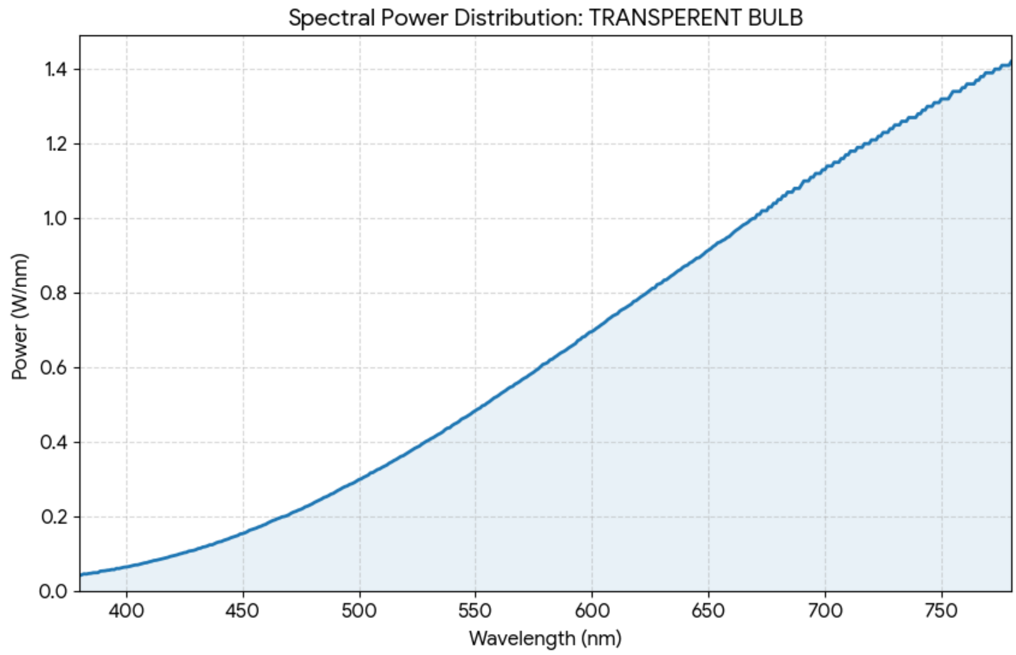


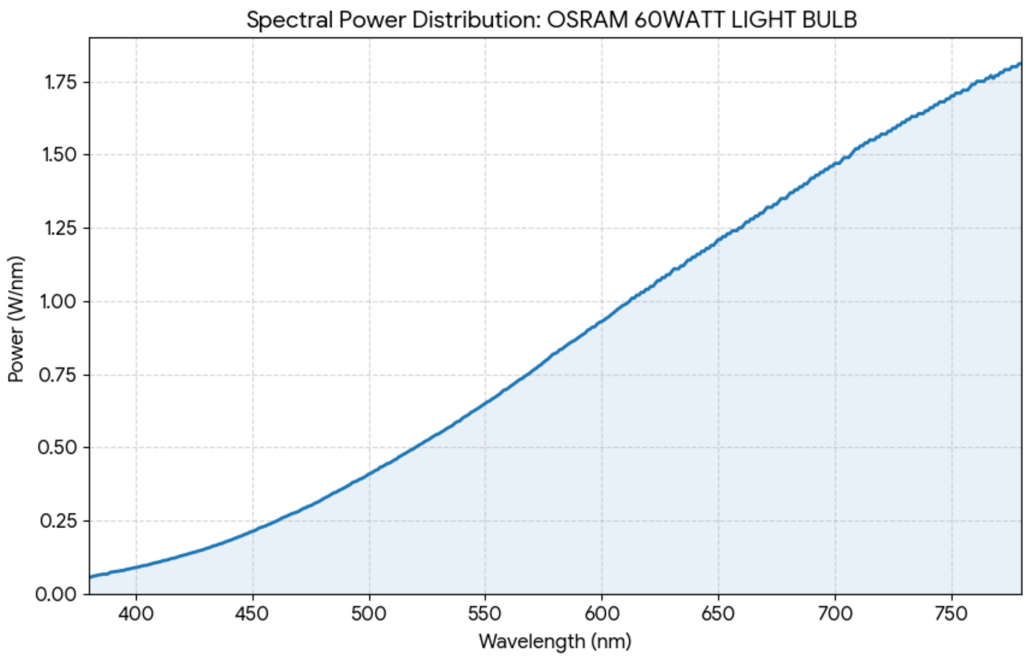


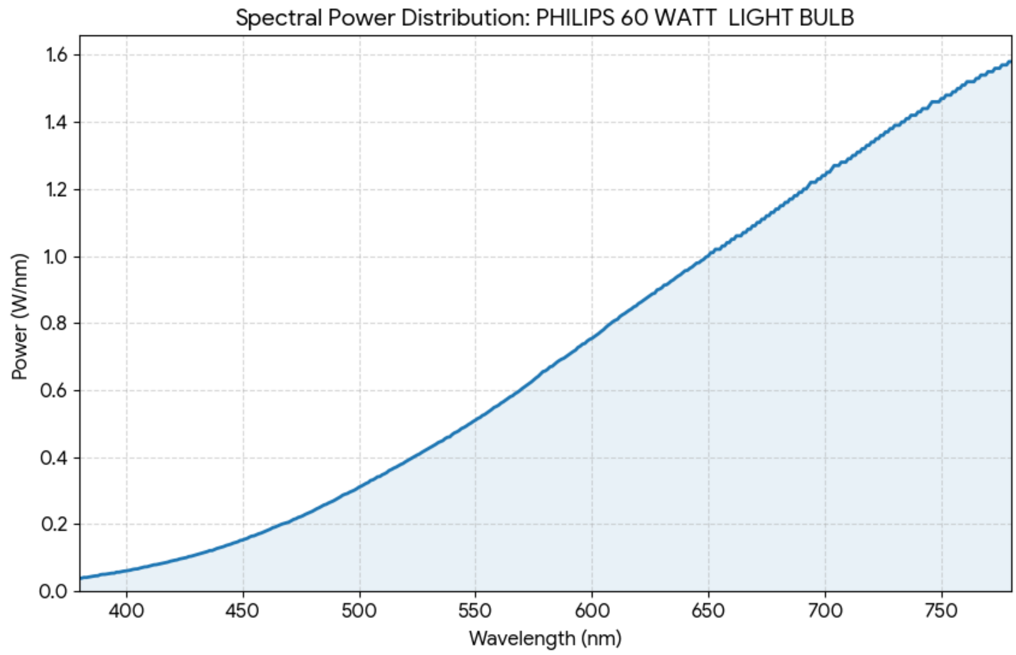
Spectral Power Distribution: ORANGE BULB
Power (W/nm)
0.00
0.05
0.10
0.15
0.20
0.25
400
450
500
550
600
650
700
750
Wavelength (nm)

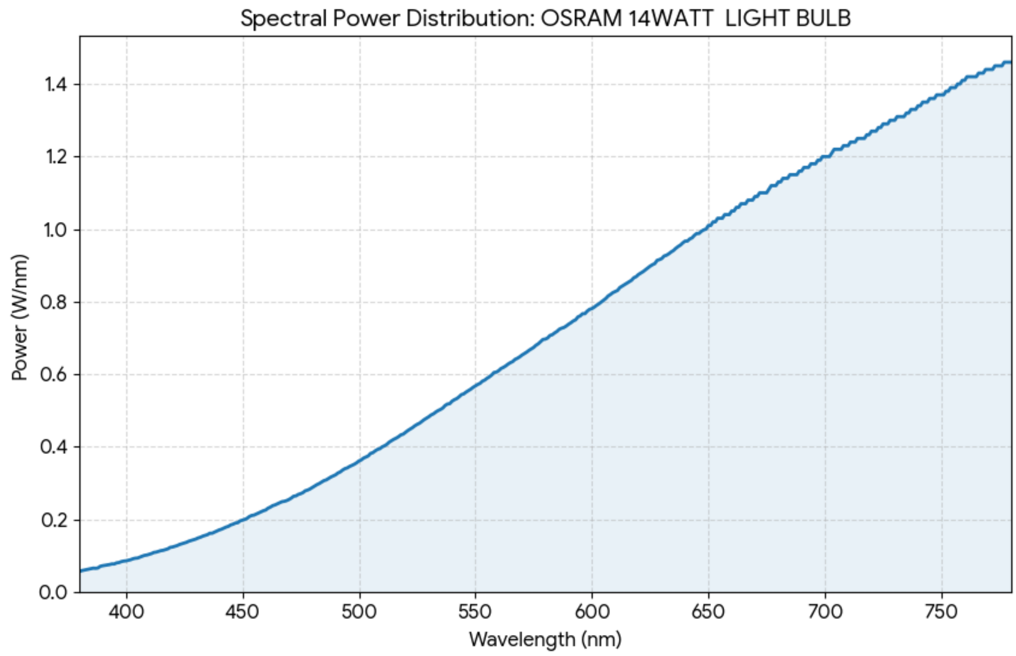


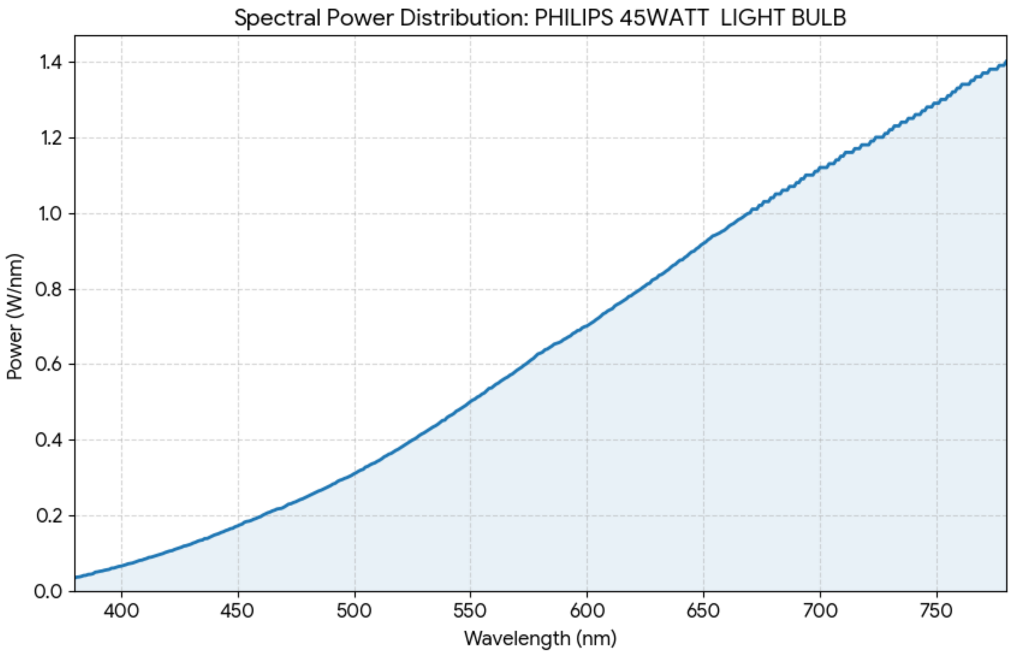
Spectral Power Distribution: RED LIGHT BULB
Power (W/nm)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
400
450
500
550
600
650
700
750
Wavelength (nm)

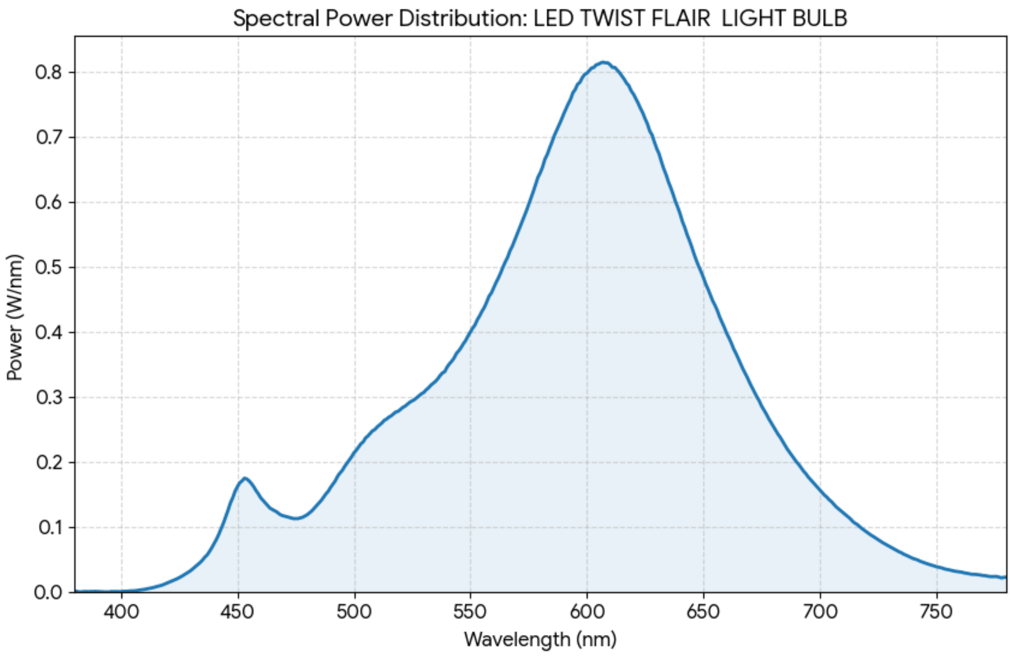

Spectral Power Distribution: TRANSPERENT BULB
Power (W/nm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
400
450
500
550
600
650
700
750
Wavelength (nm)

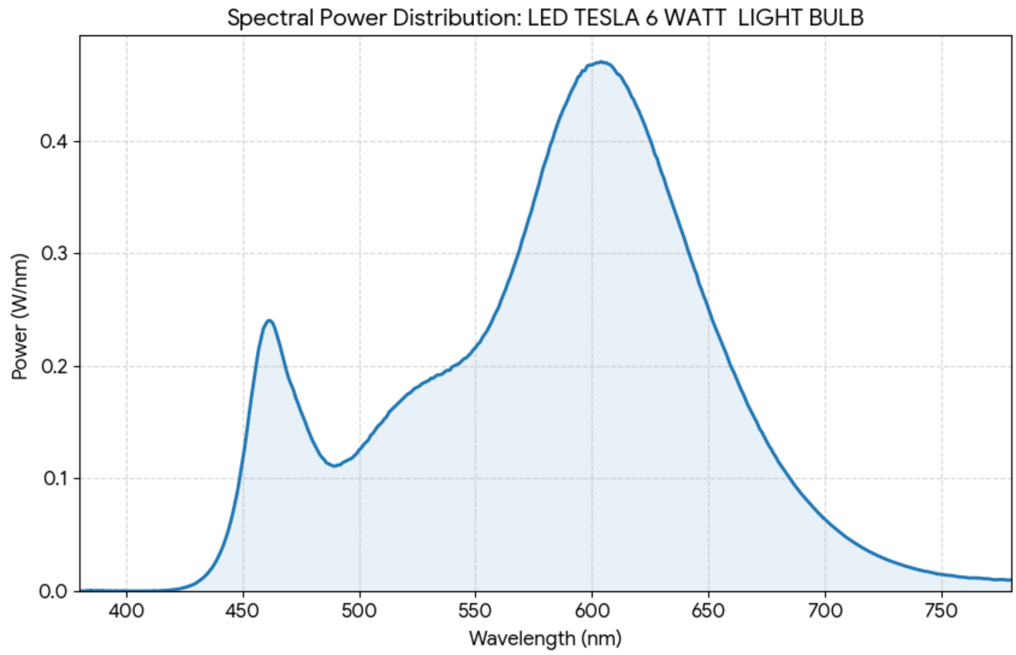

Spectral Power Distribution: OSRAM 60WATT LIGHT BULB
Power (W/nm)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
400
450
500
550
600
650
700
750
Wavelength (nm)

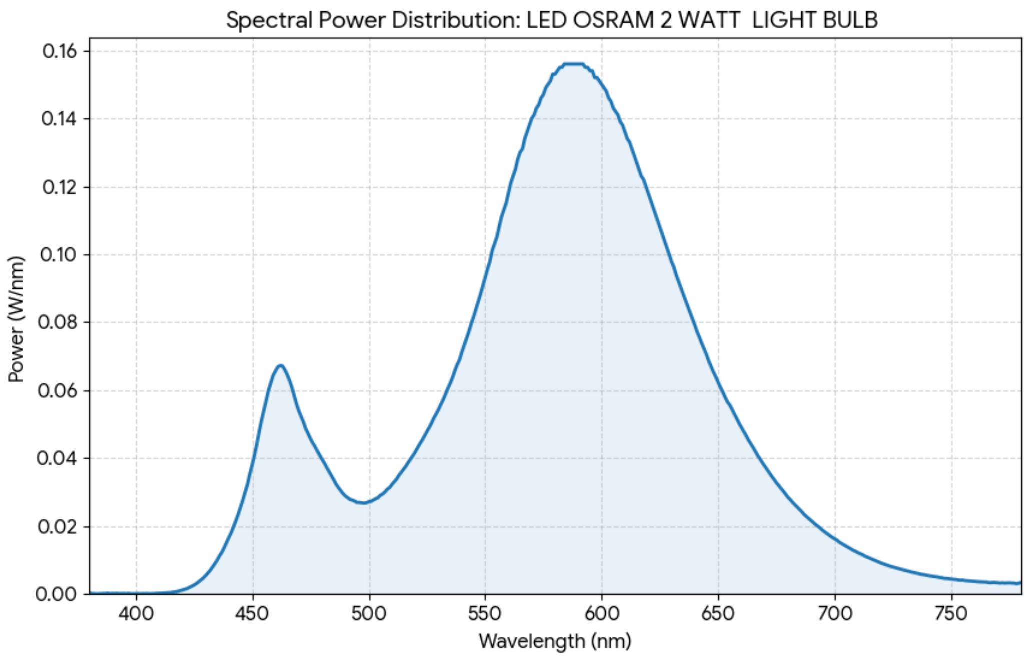

Spectral Power Distribution: PHILIPS 60 WATT LIGHT BULB
Power (W/nm)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
400
450
500
550
600
650
700
750
Wavelength (nm)

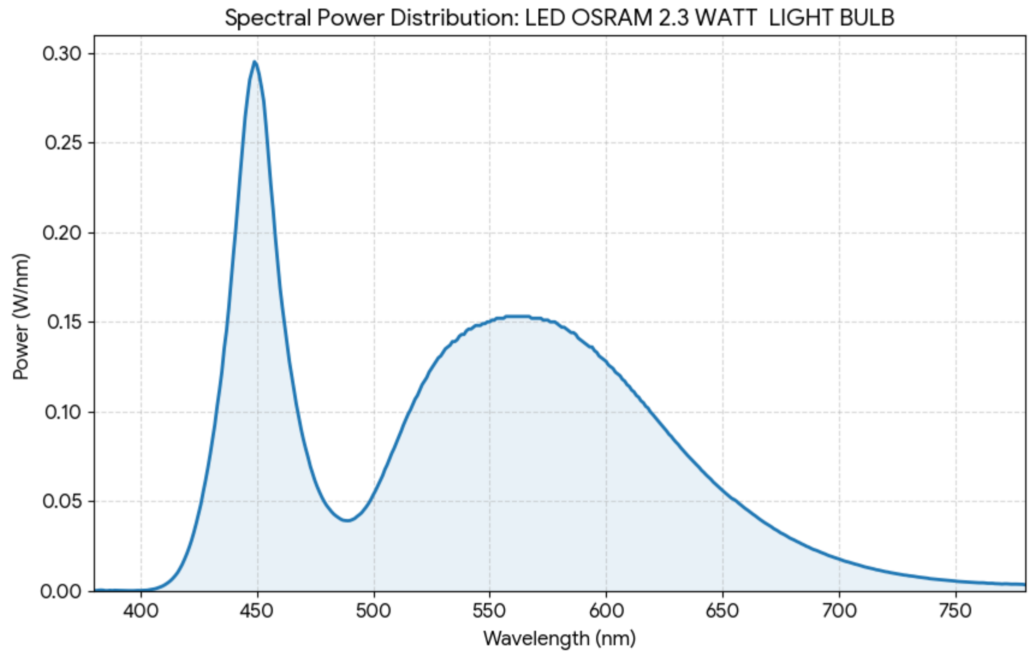

Spectral Power Distribution: OSRAM 14WATT LIGHT BULB
Power (W/nm)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
400
450
500
550
600
650
700
750
Wavelength (nm)

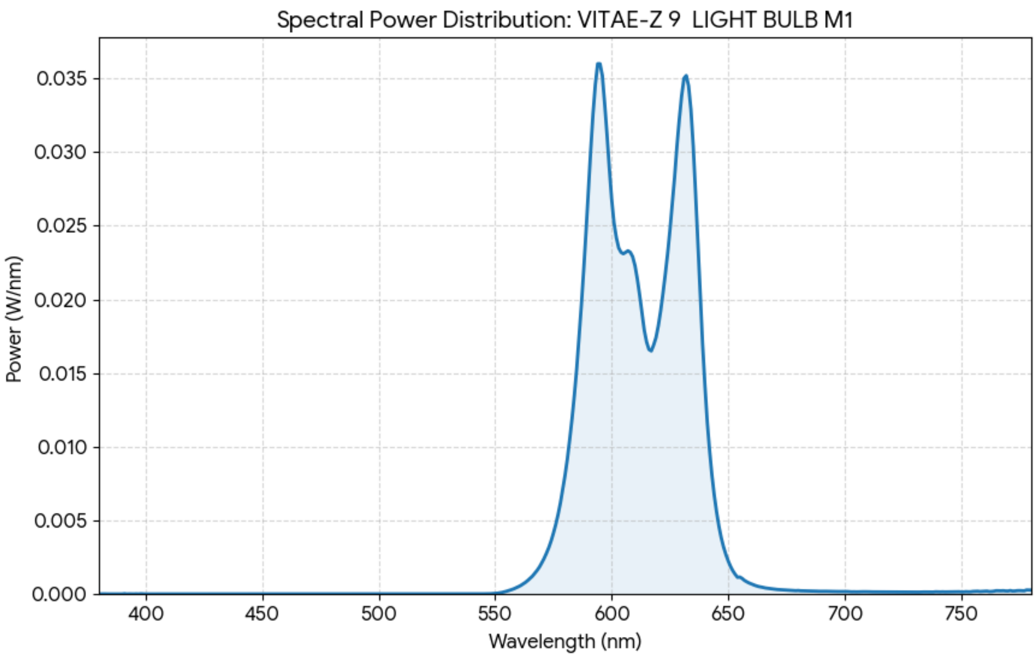

Spectral Power Distribution: PHILIPS 45WATT LIGHT BULB
Power (W/nm)
Wavelength (nm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
400
450
500
550
600
650
700
750

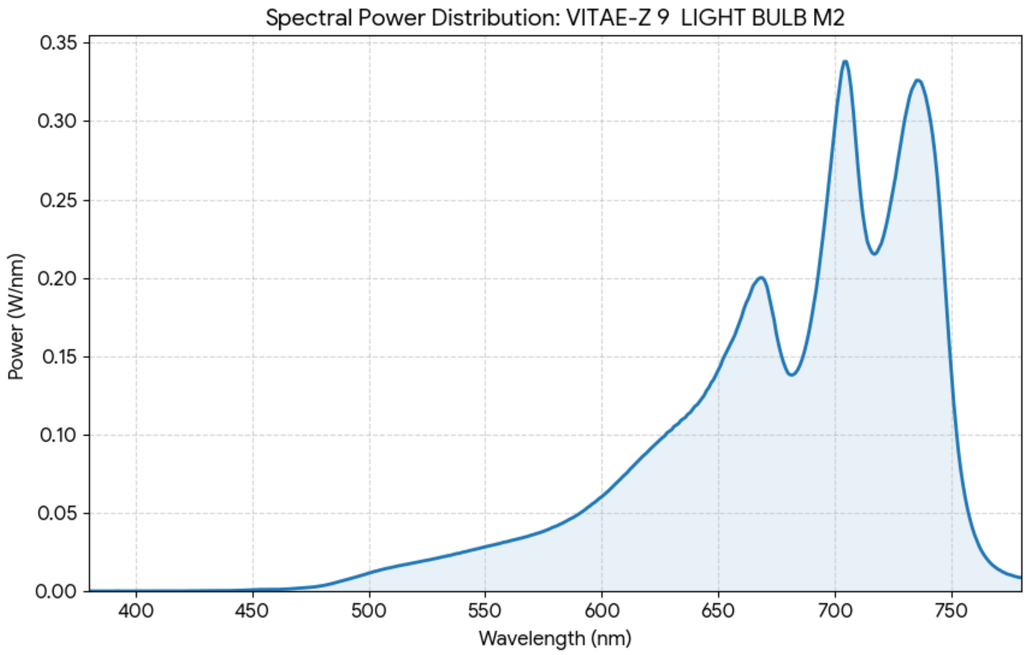

Spectral Power Distribution: LED TWIST FLAIR LIGHT BULB
Power (W/nm)
Wavelength (nm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
400
450
500
550
600
650
700
750

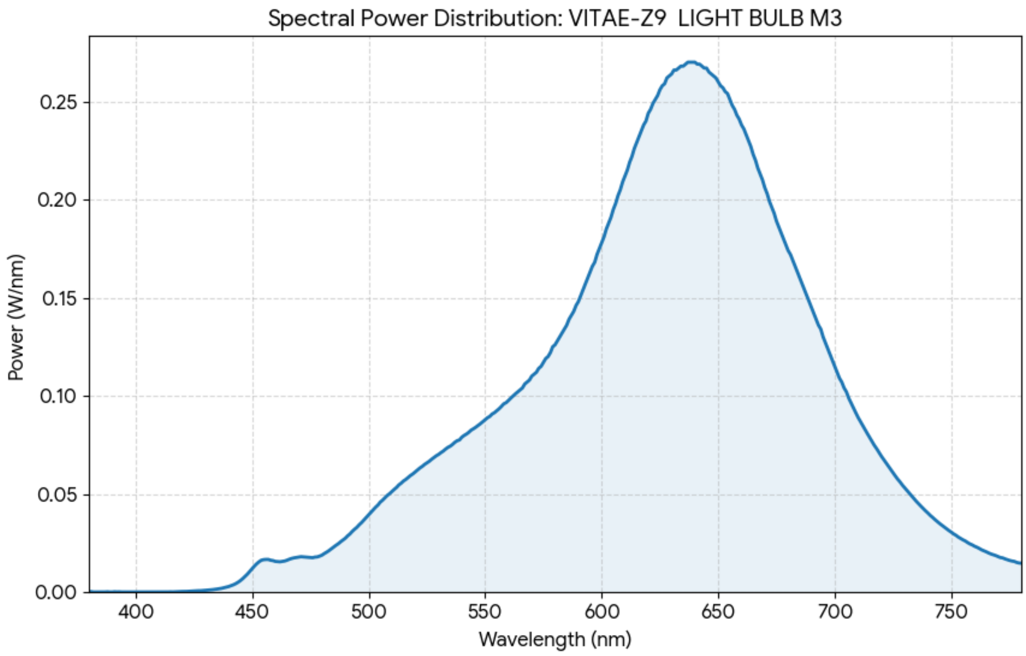

Spectral Power Distribution: LED TESLA 6 WATT LIGHT BULB
Power (W/nm)
0.0
0.1
0.2
0.3
0.4
400
450
500
550
600
650
700
750
Wavelength (nm)

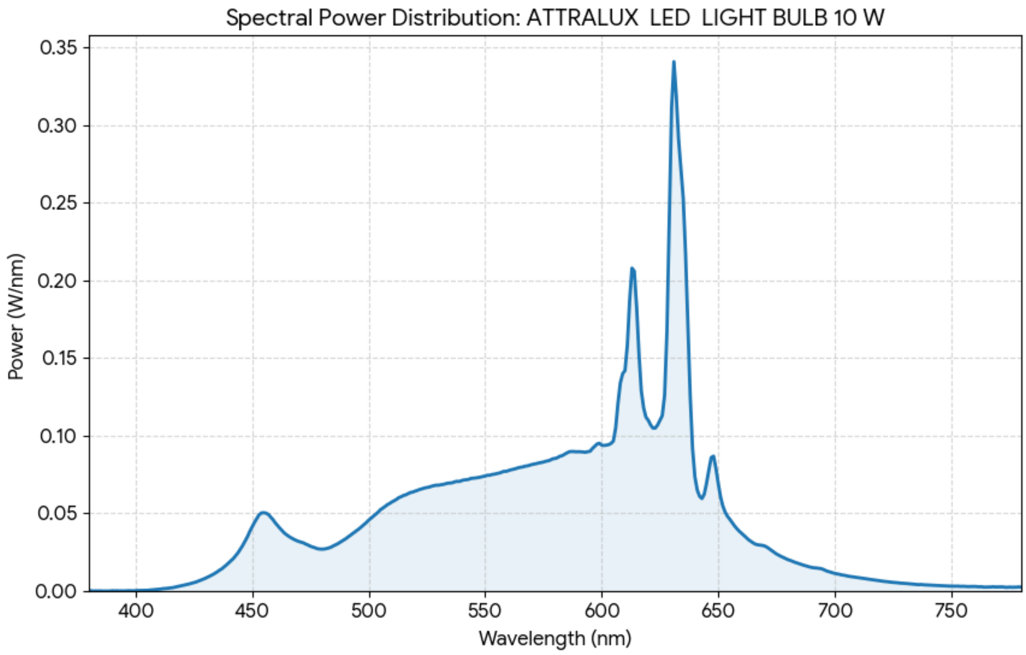

Spectral Power Distribution: LED OSRAM 2 WATT LIGHT BULB
Power (W/nm)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
400
450
500
550
600
650
700
750
Wavelength (nm)


Spectral Power Distribution: LED OSRAM 2.3 WATT LIGHT BULB
Power (W/nm)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
400
450
500
550
600
650
700
750
Wavelength (nm)

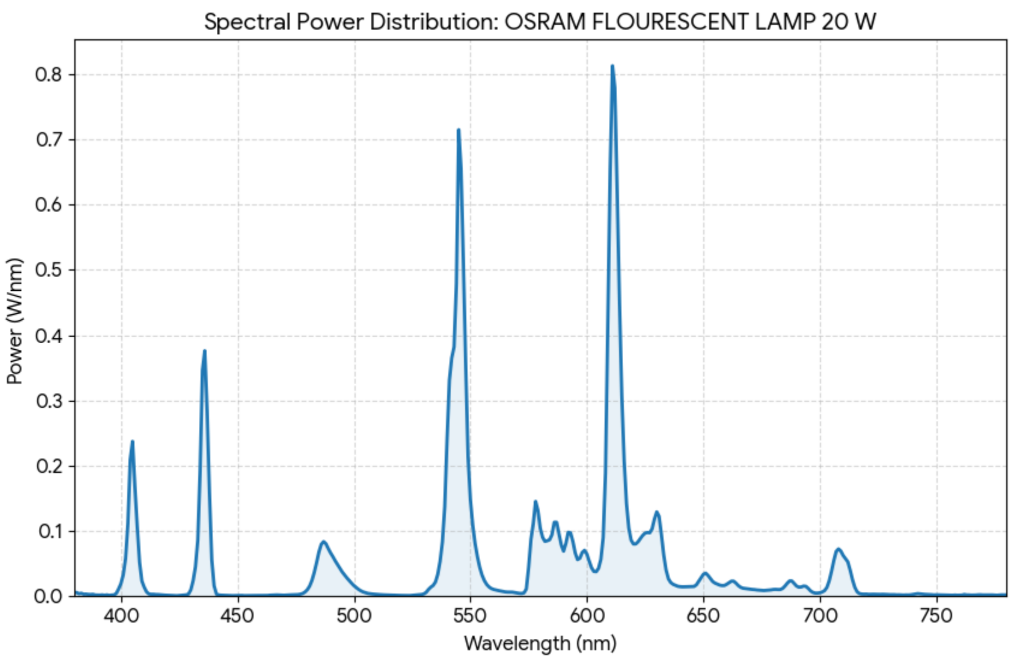

Spectral Power Distribution: VITAE-Z 9 LIGHT BULB M1
Power (W/nm)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
400
450
500
550
600
650
700
750
Wavelength (nm)


Spectral Power Distribution: VITAE-Z 9 LIGHT BULB M2
Power (W/nm)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
400
450
500
550
600
650
700
750
Wavelength (nm)

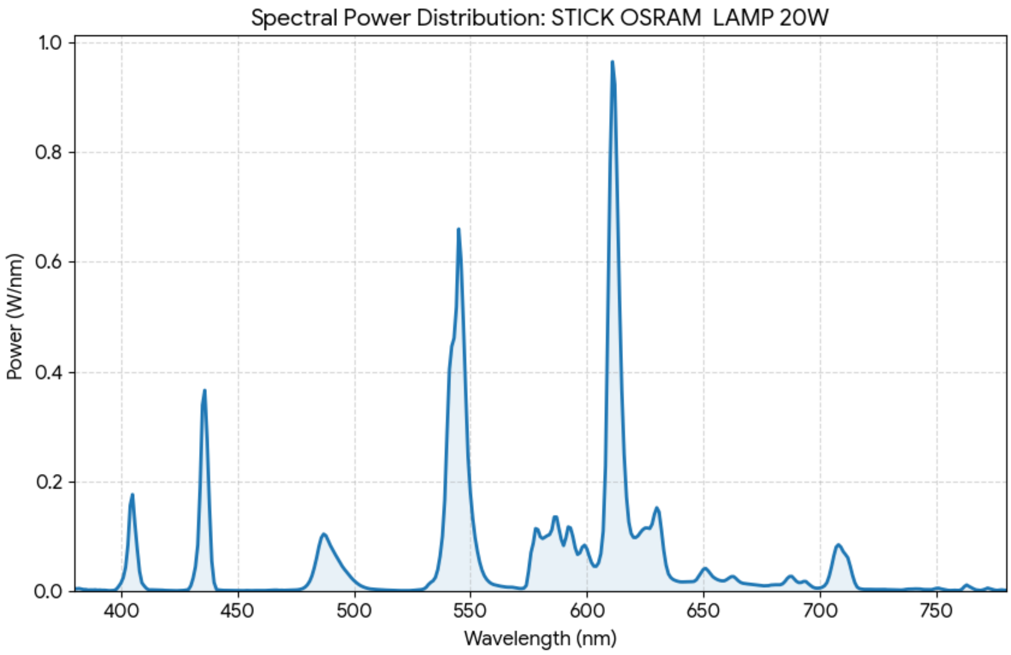

Spectral Power Distribution: VITAE-Z9 LIGHT BULB M3
Power (W/nm)
0.25
0.20
0.15
0.10
0.05
0.00
400
450
500
550
600
650
700
750
Wavelength (nm)

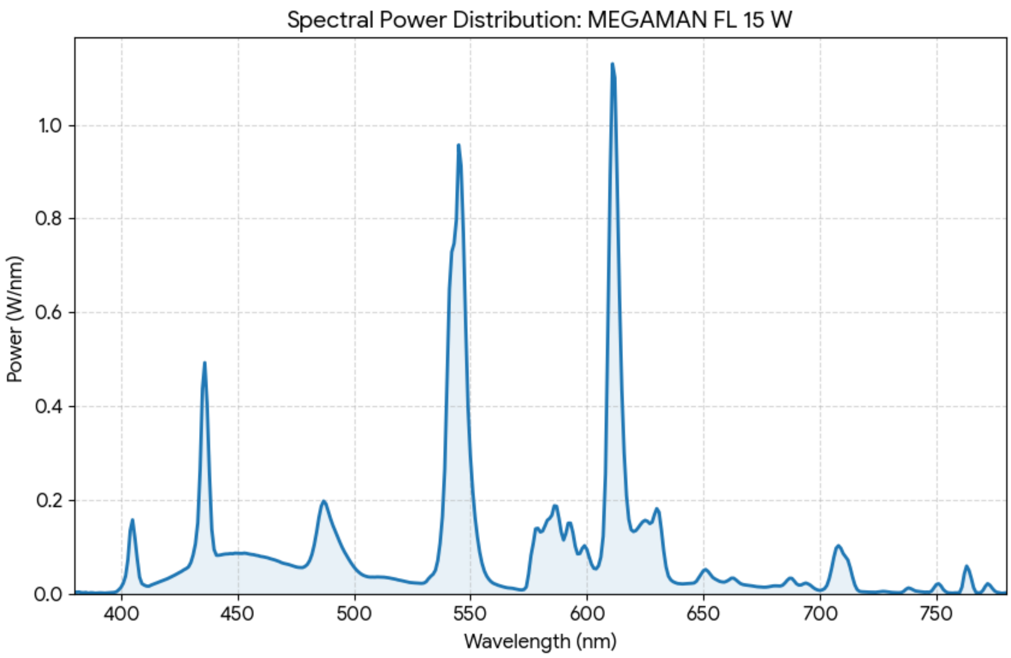


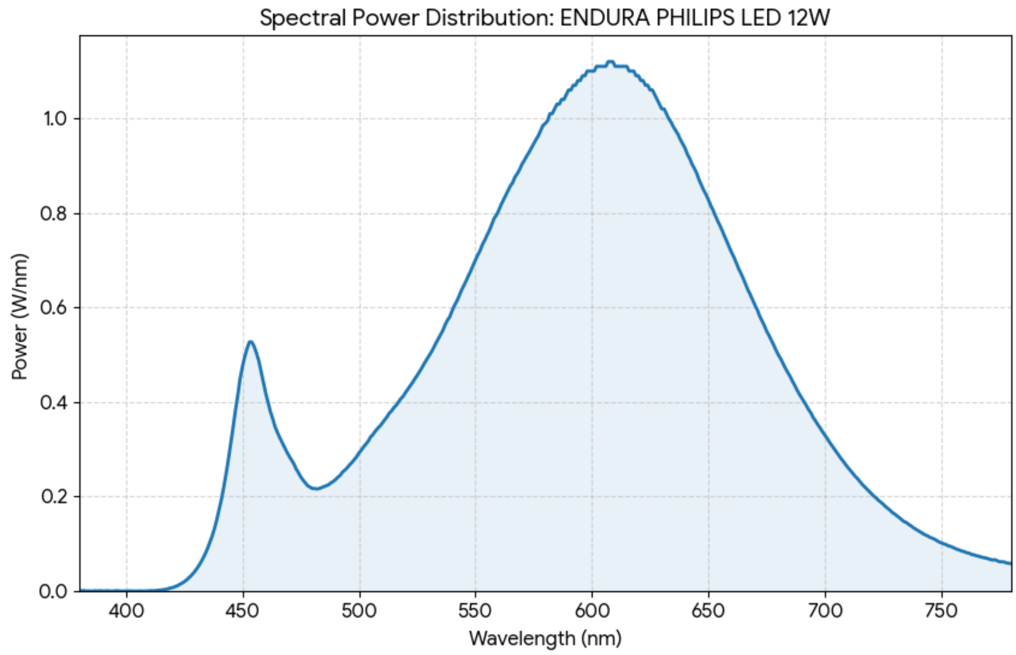
Spectral Power Distribution: ATTRALUX LED LIGHT BULB 10 W
Power (W/nm)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
400
450
500
550
600
650
700
750
Wavelength (nm)

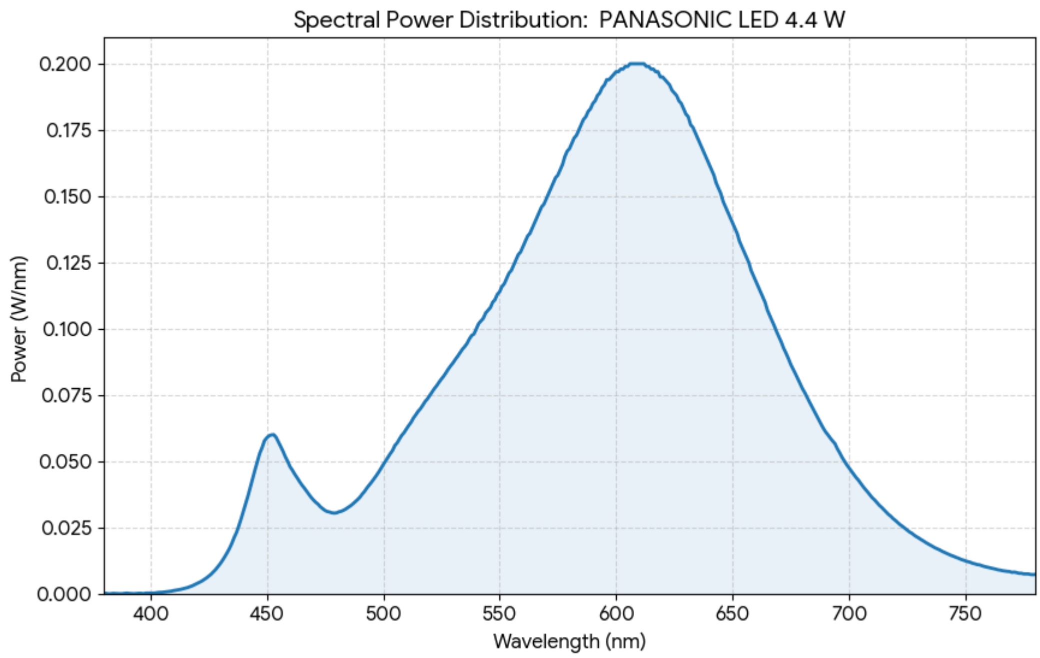


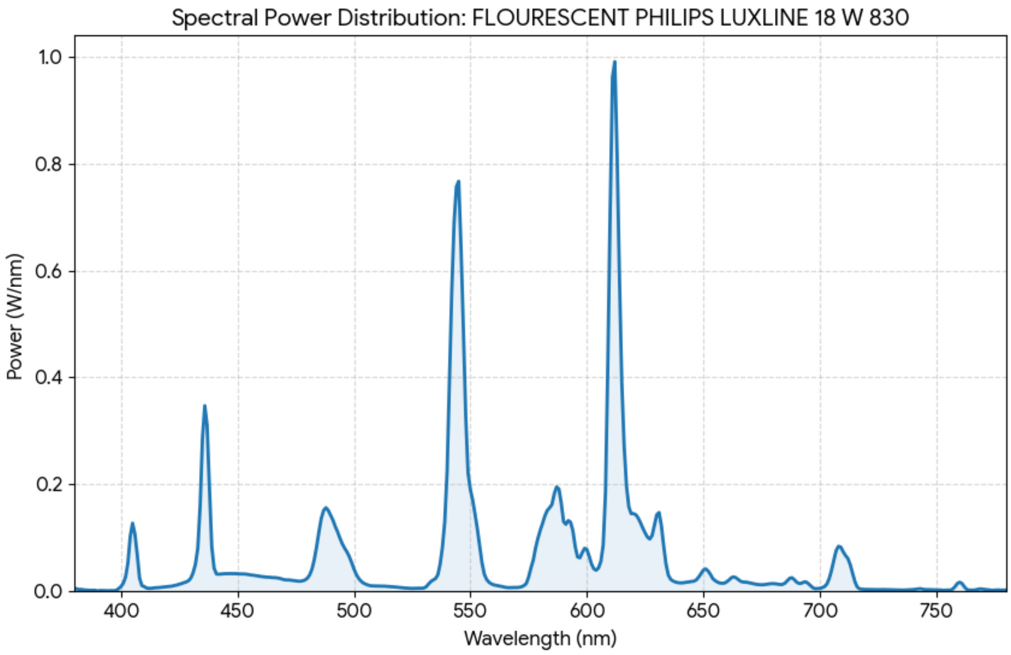
Spectral Power Distribution: OSRAM FLOURESCENT LAMP 20 W
Power (W/nm)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
400
450
500
550
600
650
700
750
Wavelength (nm)


Spectral Power Distribution: STICK OSRAM LAMP 20W
Power (W/nm)
1.0
0.8
0.6
0.4
0.2
0.0
400
450
500
550
600
650
700
750
Wavelength (nm)

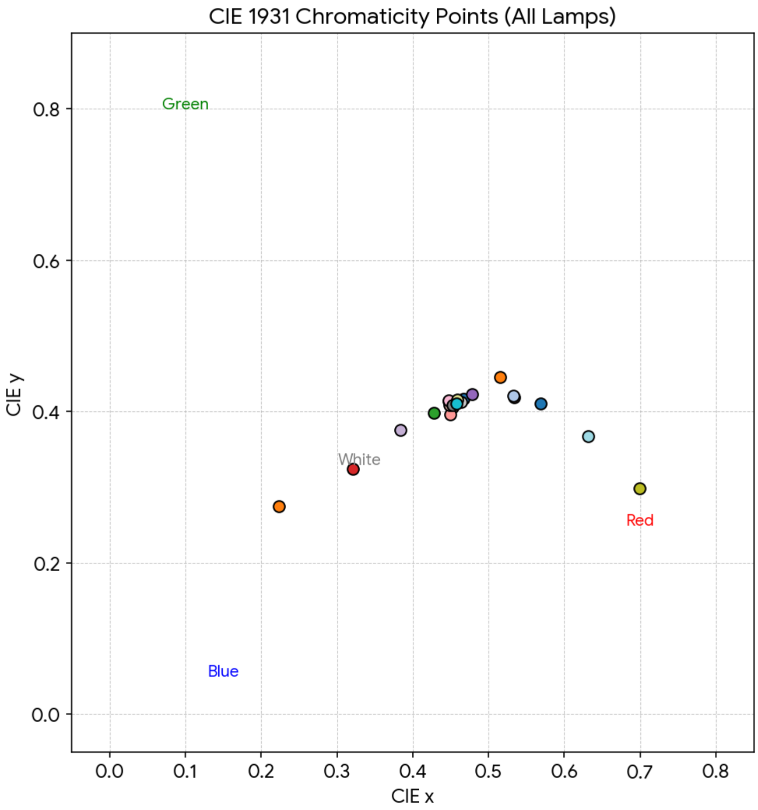

Spectral Power Distribution: MEGAMAN FL 15 W
Power (W/nm)
1.0
0.8
0.6
0.4
0.2
0.0
400
450
500
550
600
650
700
750
Wavelength (nm)


Spectral Power Distribution: ENDURA PHILIPS LED 12W
Power (W/nm)
0.0
0.2
0.4
0.6
0.8
1.0
400
450
500
550
600
650
700
750
Wavelength (nm)


Spectral Power Distribution: PANASONIC LED 4.4 W
Power (W/nm)
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
400
450
500
550
600
650
700
750
Wavelength (nm)



The colorimetric colour space x y with the chromaticity coordinates of the individual sources:



6 Conclusion

Colour perception adds a rich dimension to how we view the world, bridging the gap between physical light and our eyes' sensory receptors. While we often assume that what we see is an accurate reflection of a light source, the reality is biologically complex. Our eyes rely on three types of cone cells—tuned to short, medium, and long wavelengths—to interpret blue, green, and red light. Because these cones absorb wavelengths at different rates, our perception of a light's colour can be subjective; for example, white light might appear tinted depending on how our eyes process it.

To navigate these complexities, lighting technologists use specific metrics to quantify light. They use luminous flux to determine overall brightness, luminous intensity to track light distribution, and luminance to judge how bright a surface appears. Additionally, the "mood" of the light is defined by its Correlated Colour Temperature (CCT), distinguishing between warm and cool tones.

In short, lighting technology is about more than just physics; it is about the user experience. By mastering the relationship between illumination and human perception, we can design lighting that does more than just help us see—it can help us feel better and work more efficiently. This understanding is vital across many industries, from interior design to medicine, where lighting plays a pivotal role in human well-being.